Distributionally Robust LMI Control

Direct Data-Driven and DRC Synthesis

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Chapter 1

Distributionally Robust LMI Synthesis for LTI Systems

1.1 Problem Setting [1]

Output Feedback Control.

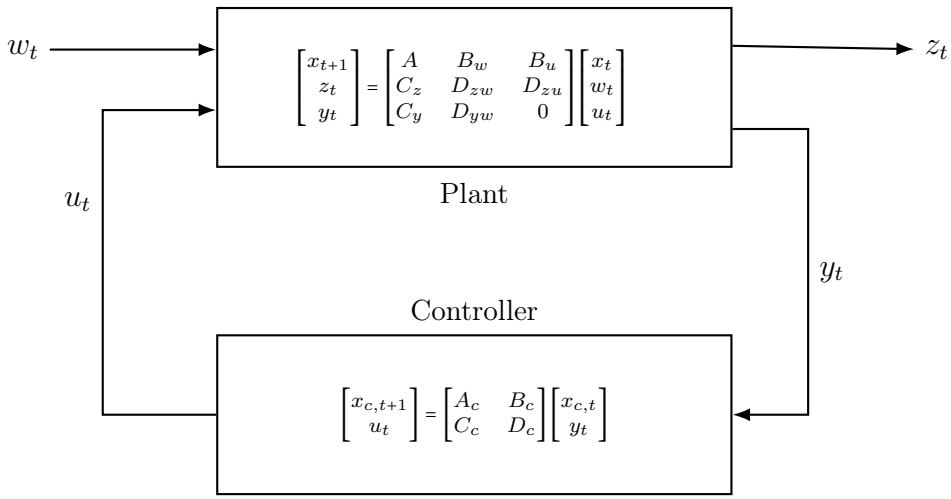


Figure 1.1: Closed-loop system with y feedback.

Define the combined state $\chi_t = [x_t^\top, x_{c,t}^\top]^\top$ and the closed-loop matrices

$$\mathbb{A} = \begin{bmatrix} A + B_u D_c C_y & B_u C_c \\ B_c C_y & A_c \end{bmatrix}, \quad \mathbb{B} = \begin{bmatrix} B_w + B_u D_c D_{yw} \\ B_c D_{yw} \end{bmatrix},$$
$$\mathbb{C} = \begin{bmatrix} C_z + D_{zu} D_c C_y & D_{zu} C_c \end{bmatrix}, \quad \mathbb{D} = D_{zw} + D_{zu} D_c D_{yw}.$$

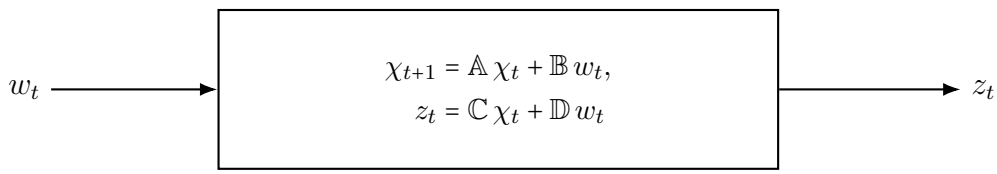


Figure 1.2: Lifted closed loop with state χ_t and matrices $(\mathbb{A}, \mathbb{B}, \mathbb{C}, \mathbb{D})$.

The disturbance formulation is formulated in A.

1.2 Paper-like Model Setting

The *paper-like* configuration [2] refers to a linear time-invariant (LTI) system of the form:

$$\begin{bmatrix} x_{t+1} \\ z_t \\ y_t \end{bmatrix} = \begin{bmatrix} A & B_w & B_u \\ C_z & D_{zw} & D_{zu} \\ C_y & D_{yw} & 0 \end{bmatrix} \begin{bmatrix} x_t \\ w_t \\ u_t \end{bmatrix},$$

where $x_t \in \mathbb{R}^{n_x}$ denotes the state, $u_t \in \mathbb{R}^{n_u}$ the control input, and $w_t \in \mathbb{R}^{n_w}$ an exogenous disturbance. The vectors $y_t \in \mathbb{R}^{n_y}$ and $z_t \in \mathbb{R}^{n_z}$ represent the measured and performance outputs, respectively.

The continuous-time version of this model describes a flexible turbine–shaft–pitch system:

$$\dot{x}(t) = A_{CT}x(t) + B_{CT}u(t) + E_{CT}w(t),$$

with

$$A_{CT} = \begin{bmatrix} \frac{k_\omega}{J} & \frac{k_h}{J} & 0 & \frac{k_\phi}{J} & 0 & \frac{k_\beta}{J} & 0 \\ 0 & 0 & 1 & 0 & 0 & 0 & 0 \\ \frac{f_\omega}{m} & -\omega_f^2 & -2\zeta_f\omega_f & 0 & 0 & \frac{f_\beta}{m} & 0 \\ 0 & 0 & 0 & 0 & 1 & 0 & 0 \\ \frac{m_\omega}{I_t} & 0 & 0 & -\omega_t^2 & -2\zeta_t\omega_t & \frac{m_\beta}{I_t} & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 1 \\ 0 & 0 & 0 & 0 & 0 & -\omega_p^2 & -2\zeta_p\omega_p \end{bmatrix},$$

$$B_{CT} = \begin{bmatrix} 0 \\ 0 \\ 0 \\ 0 \\ 0 \\ 0 \\ \omega_p^2 \end{bmatrix}, \quad E_{CT} = \begin{bmatrix} \frac{k_v}{J} & 0 \\ 0 & 0 \\ \frac{f_v}{m} & \frac{f_v}{m} \\ 0 & 0 \\ \frac{m_v}{I_t} & \frac{m_v}{I_t} \\ 0 & 0 \\ 0 & 0 \end{bmatrix}.$$

which is discretized with sampling time T_s via

$$A = e^{A_{CT}T_s}, \qquad B_u = \int_0^{T_s} e^{A_{CT}\tau} B_{CT}\, d\tau, \qquad B_w = \int_0^{T_s} e^{A_{CT}\tau} E_{CT}\, d\tau.$$

The output matrices $(C_z, D_{zw}, D_{zu}, C_y, D_{yw})$ are selected to reflect a standard $\mathcal{H}_2$-type performance objective:

$$z_t = \begin{bmatrix} Q_x^{1/2} & 0 \\ 0 & R_u^{1/2} \end{bmatrix} \begin{bmatrix} x_t \\ u_t \end{bmatrix}, \qquad Q_x \succeq 0, R_u \succ 0 \qquad y_t = C_y x_t + D_{yw} w_t,$$

with $Q_x \succeq 0, R_u \succ 0$ and $D_{zw} = 0$.

1.3 Optimisation Problems [1]

Baseline

$$\min_{\theta \leftrightarrow (A_c, B_c, C_c, D_c)} J(\theta), \qquad \text{s.t. } \mathbb{A} \quad \text{isstable}$$

$$J(\theta) = \sup_{w \in \mathcal{W}} \quad \limsup_{T \to \infty} \frac{1}{T} \sum_{t=0}^{T-1} \mathbb{E} \, \|z_t\|^2 \tag{P1}$$

$$\text{s.t.} \quad \begin{aligned} \chi_{t+1} &= \mathbb{A}\,\chi_t + \mathbb{B}\,w_t, \\ z_t &= \mathbb{C}\,\chi_t + \mathbb{D}\,w_t \end{aligned}, \quad \chi_0 = 0$$

LMI

Introducing decision variables

$$\lambda \geq 0, \quad Q, \quad X = X^\top, \; Y = Y^\top, \; K, L, M, N,$$

and defining

$$\mathbb{P} = \begin{bmatrix} Y & I \\ I & X \end{bmatrix}, \qquad \mathbf{A} = \begin{bmatrix} AY + B_u M & A + B_u N C_y \\ K & XA + LC_y \end{bmatrix},$$

$$\mathbf{B} = \begin{bmatrix} B_w + B_u N D_{yw} \\ XB_w + LD_{yw} \end{bmatrix}, \quad \mathbf{C} = \begin{bmatrix} C_z Y + D_{zu} M & C_z + D_{zu} N C_y \end{bmatrix}, \quad \mathbf{D} = D_{zw} + D_{zu} N D_{yw}.$$

The objective bacame:

$$\min_{\lambda \geq 0, \; Q, \; X = X^\top, \; Y = Y^\top, \; K, L, M, N} \quad \operatorname{trace}(Q \Sigma_{\text{nom}}) + \lambda \gamma^2 \tag{P2}$$

subject to a block LMI that depends on the ambiguity model.

Correlated model ($\mathcal{W}_{\text{cor}}$). With identity blocks I_w, I_z of matching sizes, enforce

$$\begin{bmatrix} -\mathbb{P} & 0 & 0 & \mathbf{A}^\top & \mathbf{C}^\top \\ 0 & -\lambda I_w & \lambda I_w & \mathbf{B}^\top & \mathbf{D}^\top \\ 0 & \lambda I_w & -Q - \lambda I_w & 0 & 0 \\ \mathbf{A} & \mathbf{B} & 0 & -\mathbb{P} & 0 \\ \mathbf{C} & \mathbf{D} & 0 & 0 & -I_z \end{bmatrix} \prec 0, \quad \mathbb{P} \succeq 0.$$

Independent model ($\mathcal{W}_{\text{ind}}$). Impose the pair of LMIs

$$\begin{bmatrix} -\mathbb{P} & \mathbf{A}^\top & \mathbf{C}^\top \\ \mathbf{A} & -\mathbb{P} & 0 \\ \mathbf{C} & 0 & -I_z \end{bmatrix} \prec 0, \qquad \begin{bmatrix} -\lambda I_w & \lambda I_w & \mathbf{B}^\top & \mathbf{D}^\top \\ \lambda I_w & -Q - \lambda I_w & 0 & 0 \\ \mathbf{B} & 0 & -\mathbb{P} & 0 \\ \mathbf{D} & 0 & 0 & -I_z \end{bmatrix} \prec 0, \quad \mathbb{P} \succeq 0.$$

Chapter 2

Data-Driven Approach

We estimate the plant

$$\mathcal{P} = (A, B_u, B_w, C_z, D_{zw}, D_{zu}, C_y, D_{yw})$$

directly from a single trajectory $\{x_t, u_t, y_t, z_t\}_{t=0}^{T-1}$ by aligned one-step regression. Throughout, vectors are column-valued, all sample matrices are column-stacked (variables by time), and expectations are replaced by empirical averages (with ridge terms to tame conditioning).

2.1 Preprocessing and Notation

Blocks. Build aligned blocks over a common horizon T:

$$\begin{aligned}
&X \in \mathbb{R}^{n_x \times T} && X := [\, x_0 \; \dots \; x_{T-1} \,], \\
&U \in \mathbb{R}^{n_u \times T} && U := [\, u_0 \; \dots \; u_{T-1} \,], \\
&X^+ \in \mathbb{R}^{n_x \times T} && X^+ := [\, x_1 \; \dots \; x_T \,], \\
&Y \in \mathbb{R}^{n_y \times T}, && Z \in \mathbb{R}^{n_z \times T} \text{ if available.}
\end{aligned}$$

Define the concatenated regressor $D := \begin{bmatrix} X \\ U \end{bmatrix} \in \mathbb{R}^{(n_x+n_u)\times(T-1)}$.

Persistency. We require the empirical persistency-of-excitation (PE) condition

$$rk(D) = n_x + n_u,$$

else the identification is ill-posed.

2.2 State and Input Matrices (A, B_u)

Ridge-regularized least squares gives the minimum-variance linear predictor for x_{t+1} given (x_t, u_t):

$$\boxed{\begin{bmatrix} \widehat{A} & \widehat{B}_u \end{bmatrix} = X^+ D^\top \left(DD^\top + \rho_x I_{n_x+n_u} \right)^{-1},} \qquad \rho_x > 0. \tag{2.1}$$

Shapes: $DD^\top \in \mathbb{R}^{(n_x+n_u)\times(n_x+n_u)}$.

Residual construction. Once (A, B_u) are estimated by regression, the remaining unexplained dynamics at each step

$$R := X^+ - A\,X - B_u\,U \in \mathbb{R}^{n_x \times T}$$

represent the *innovation sequence*—the portion of the next state not predicted by (x_t, u_t).

In the linear–Gaussian model with arbitrary covariance

$$x_{t+1} = Ax_t + B_u u_t + B_w w_t, \qquad w_t \sim \mathcal{N}(0, \Sigma_w), \quad \Sigma_w \succ 0,$$

the residuals satisfy (in expectation)

$$\mathbb{E}\left[\frac{1}{T} RR^\top\right] = B_w \Sigma_w B_w^\top.$$

Thus, the empirical residual covariance

$$S_R := \frac{1}{T} RR^\top \;\approx\; B_w \Sigma_w B_w^\top$$

estimates the disturbance covariance as seen through the map B_w.

Covariance decomposition. Define the residual covariance matrix

$$S_R := \frac{1}{T-1} RR^\top = U_R\,\Sigma_R\,U_R^\top,$$

with eigenvalues $\Sigma_R = \mathrm{diag}(\sigma_1 \geq \cdots \geq \sigma_{n_x} \geq 0)$. Each σ_i represents the energy of a principal disturbance direction, evaluated via SVD (Singular Value Decomposition).

Rank truncation and energy criterion. Noise and modeling imperfections make S_R full-rank even if the true B_w has low dimension $n_w < n_x$. To recover a parsimonious model, retain the smallest k such that

$$\frac{\sum_{i=1}^{k} \sigma_i}{\sum_{i=1}^{n_x} \sigma_i} \;\geq\; \eta, \qquad \eta \in (0, 1] \text{ (e.g. } \eta = 0.95).$$

Then construct

$$\boxed{\widehat{B}_w\,\widehat{\Sigma}_w^{1/2} := U_R(:, 1{:}k)\;\Sigma_R(1{:}k, 1{:}k)^{1/2} \;\in\; \mathbb{R}^{n_x \times k},} \qquad n_w := k. \tag{2.2}$$

This ensures $B_w B_w^\top$ reproduces the dominant k-dimensional subspace of S_R and that the retained modes account for a fraction η of the total residual variance.

$$\textit{White-normalized convention: } \widehat{\Sigma}_w = I_k, \;\; \Rightarrow \;\; \widehat{B}_w = U_R(:, 1{:}k)\;\Sigma_R(1{:}k, 1{:}k)^{1/2}.$$

Interpretation. B_w provides a low-dimensional basis for the "innovation" space: each column of B_w corresponds to one independent disturbance channel driving the state dynamics. The scaling by $\Sigma_R^{1/2}$ gives physically meaningful amplitudes, so that $\mathrm{Cov}(B_w w_t) = B_w \Sigma_w B_w^\top \approx S_R$.

2.3 Output Channels (C_y, D_{yw}) and Performance Map (C_z, D_{zu}, D_{zw})

When Y and/or Z are available, perform ridge regressions that explicitly separate the state pathway and the residual pathway.

Measured output. Model $y_t = C_y x_t + D_{yw} w_t$ with w_t proxied by residuals. Stacking columns:

$$\boxed{Y \approx \underbrace{C_y}_{n_y \times n_x} X + \underbrace{D_{yw}}_{n_y \times n_w} R_w, \qquad R_w := B_w^{\dagger} R,}$$

and solve the ridge problem

$$\min_{C_y, D_{yw}} \left\| Y - [\, C_y \;\; D_{yw} \,] [\, X; R_w \,] \right\|_F^2 + \rho_y \| [\, C_y \;\; D_{yw} \,] \|_F^2,$$

which yields the closed form

$$[\, \widehat{C}_y \;\; \widehat{D}_{yw} \,] = Y \Theta_Y^{\top} (\Theta_Y \Theta_Y^{\top} + \rho_y I)^{-1}, \quad \Theta_Y := [\, X; R_w \,].$$

Performance output. Model $z_t = C_z x_t + D_{zu} u_t + D_{zw} w_t$ with the same proxy:

$$\boxed{Z \approx C_z X + D_{zu} U + D_{zw} R_w,}$$

and solve

$$[\, \widehat{C}_z \;\; \widehat{D}_{zu} \;\; \widehat{D}_{zw} \,] = Z \Theta_Z^{\top} (\Theta_Z \Theta_Z^{\top} + \rho_z I)^{-1}, \quad \Theta_Z := [\, X; U; R_w \,].$$

2.4 DD simulations

Considering the same simulation done in the paper [1], we consider *correlated* distrubance, set the nominal noise covariance to the Identity matrix and $\gamma = 0.5$, and we run the problem both in Model-Based and Data-Driven.

Reproducibility and implementation consistency

To verify the reproducibility of the results presented in [1], the same optimization problem was re-implemented in the current framework (`main.py`) and compared numerically against the reference implementation provided in `DRControllerSynthesis`. Table 2.1 reports the metrics obtained from both runs.

Table 2.1: Comparison between reference (article) and reproduced implementation.

Metric	**DRControllerSynthesis**	**main.py**	Δ
Objective value	878.6000	878.5714	$+2.86 \times 10^{-2}$
γ	0.5000	0.5000	0.00
λ_{opt}	1164.0231	1164.0234	-2.68×10^{-4}
$\rho(A_{\text{cl}})$	0.992742	0.992721	$+2.08 \times 10^{-5}$

The discrepancies between the original code and the current reimplementation are negligible, all within 10^{-4} relative precision. This confirms that the solver setup, numerical conditioning, and problem formulation have been faithfully reproduced. The small differences in the objective and spectral radius stem from finite-precision effects in the SDP solver and floating-point accumulation order. Consequently, the reproduced pipeline can be regarded as numerically equivalent to the one used in the reference article, ensuring full comparability of the subsequent simulations and results.

Results and comparison

Both Model-Based (MBD) and Data-Driven (DD) controllers converge to optimal solutions under correlated disturbances with a common ambiguity radius

$$\gamma = 0.5.$$

MOSEK reports *optimal* termination and feasibility for both problems. The closed loops are asymptotically stable, with spectral radii

$$\rho(A_{\mathrm{cl}}^{\mathrm{MBD}}) = 0.9927, \qquad \rho(A_{\mathrm{cl}}^{\mathrm{DD}}) = 0.9746.$$

LMI objectives. The optimal objective values recorded by the solver are

$$J_{\mathrm{MBD}} = 878.5714, \qquad J_{\mathrm{DD}} = 951.0618,$$

Controller and plant deviations. Both controllers share dimensions $(7, 7, 3, 1)$ for (A_c, B_c, C_c, D_c). The Frobenius norms of the deviations $\Delta = \mathrm{DD} - \mathrm{MBD}$ are

$$\|\Delta A_c\|_F = 3.2274, \quad \|\Delta B_c\|_F = 1.7388 \times 10^4, \quad \|\Delta C_c\|_F = 5.75 \times 10^{-3}, \quad \|\Delta D_c\|_F = 9.9504.$$

Plant deviations are much smaller:

$$\|\Delta A\|_F = 5.6189, \quad \|\Delta B_u\|_F = 1.5885 \times 10^{-2}, \quad \|\Delta C_y\|_F \approx 1.5 \times 10^{-8}, \quad \|\Delta C_z\|_F \approx 1.03 \times 10^{-5}.$$

In the composite closed loop,

$$\|\Delta A_{\mathrm{cl}}\|_F = 1.7388 \times 10^4, \qquad \|\Delta C_{\mathrm{cl}}\|_F = 9.9504,$$

showing that differences are concentrated at the control layer and propagate strongly into the lifted dynamics.

Conditioning. Both closed-loop matrices are full rank (14×14). Reported 2-norm condition numbers are

$$\kappa_2(A_{\mathrm{cl}}^{\mathrm{MBD}}) \approx 5.66 \times 10^8, \qquad \kappa_2(A_{\mathrm{cl}}^{\mathrm{DD}}) \approx 2.53 \times 10^{10}.$$

DD has a smaller spectral radius but is notably more ill-conditioned.

Instantaneous and time-averaged cost. The instantaneous quadratic cost $z(t)^\top z(t)$ is shown in Fig. 2.1. The time-average cost

$$J(t) = \frac{1}{t}\sum_{k=0}^{t} \|z(k)\|_2^2$$

(Fig. 2.2) converges to numerically distinct levels:

$$\overline{J}_{\mathrm{MBD}} \approx 12.2939, \qquad \overline{J}_{\mathrm{DD}} \approx 24.3136,$$

confirming that MBD achieves the lower long-horizon cost in simulation, consistent with the LMI objectives.

Time-domain KPIs. RMS and L_2 energies over the simulated horizon reflect the same pattern:

	rms(x)	rms(x_c)	rms(u)	rms(z)
MBD	1.3356	2.4261×10^3	0.6339	1.2693
DD	1.7299	2.2950×10^3	1.1116	1.6652

DD uses stronger actuation and yields larger z-RMS and cost; MBD attains lower regulation energy for z with milder inputs.

Visualisation. Figures 2.3–2.6 compare inputs, states, and outputs. The DD design exhibits faster spectral decay but with higher input effort and larger performance-output energy, while MBD attains lower cost levels overall.

Summary. Under the same ambiguity radius $\gamma = 0.5$ and correlated disturbances, the model-based solution wins on both the convex objective and the simulated time-average cost. The data-driven controller achieves a smaller spectral radius but pays with higher cost and worse conditioning.

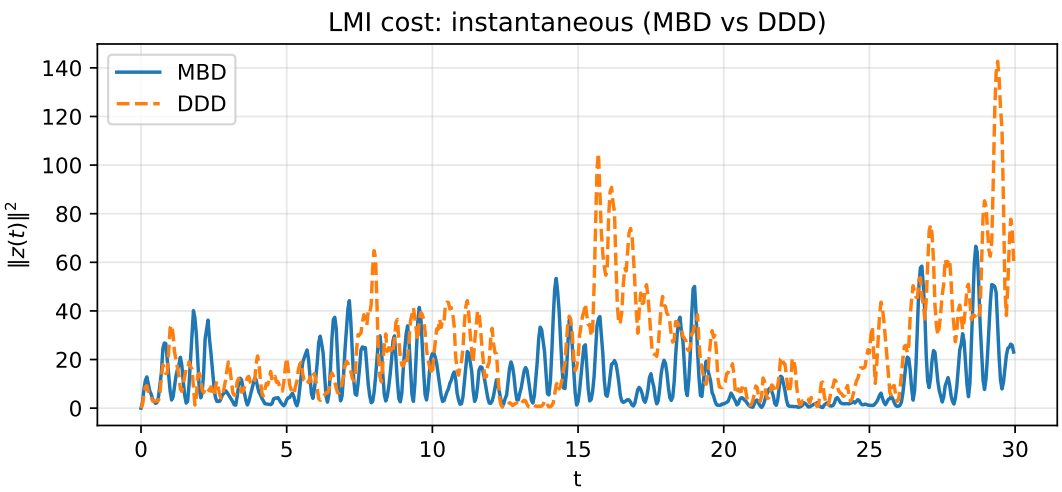


Figure 2.1: Instantaneous quadratic cost $z(t)^\top z(t)$ (MBD vs DD).

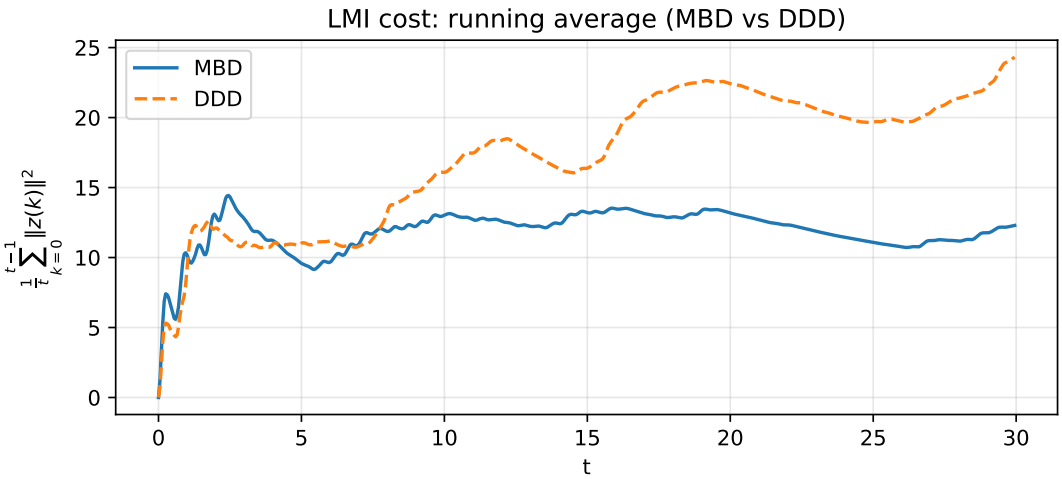


Figure 2.2: Running-average cost $\frac{1}{t}\sum_{k=0}^{t}\|z(k)\|_2^2$ (MBD vs DD). MBD converges to a lower level (≈ 12.29) than DD (≈ 24.31).

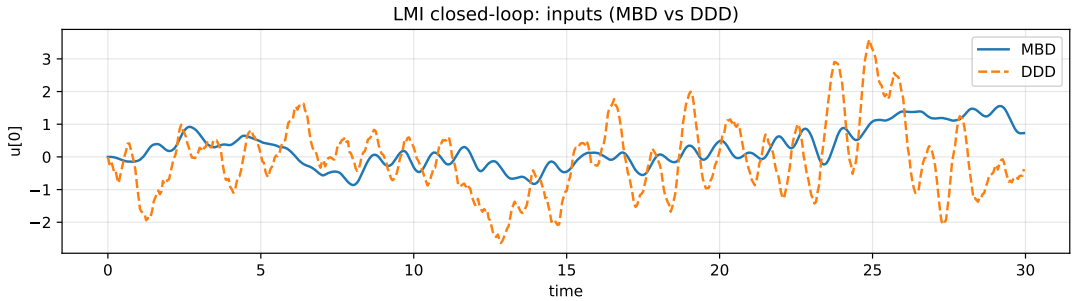


Figure 2.3: Control inputs (MBD vs DD).

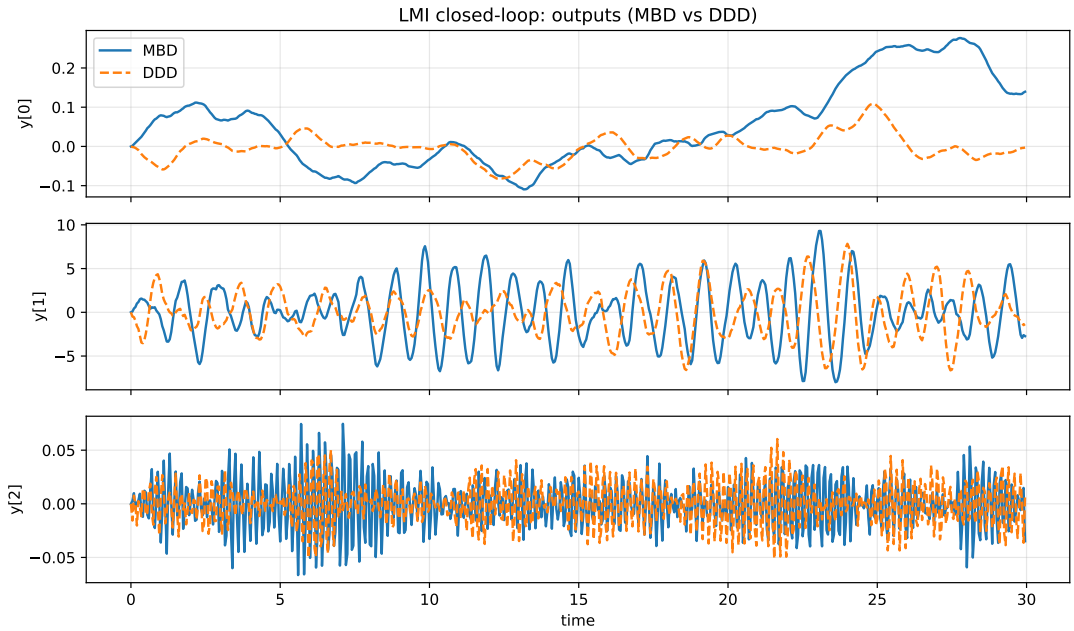


Figure 2.4: Measured outputs $y(t)$ (MBD vs DD).

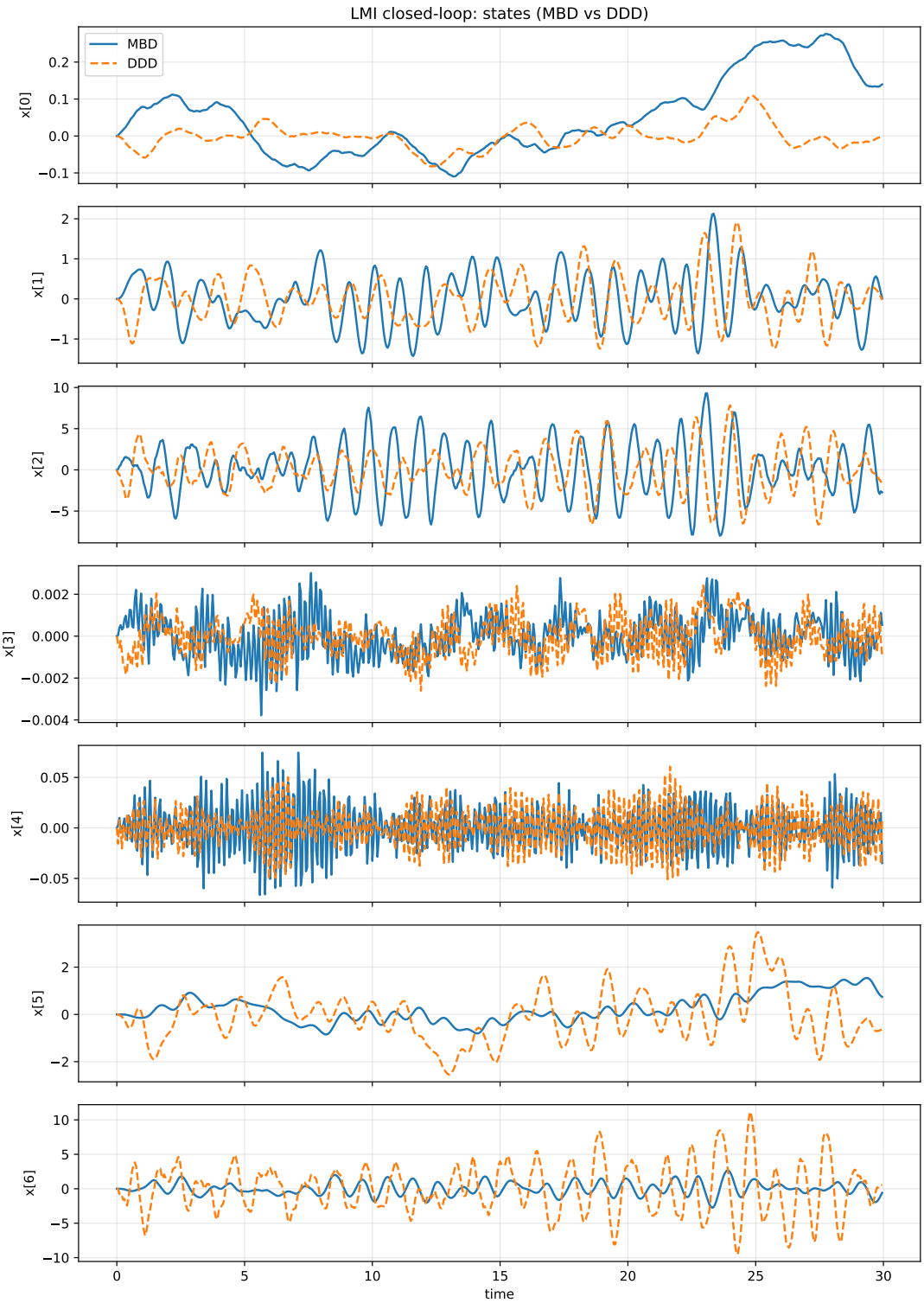


Figure 2.5: System states $x(t)$ (MBD vs DD).

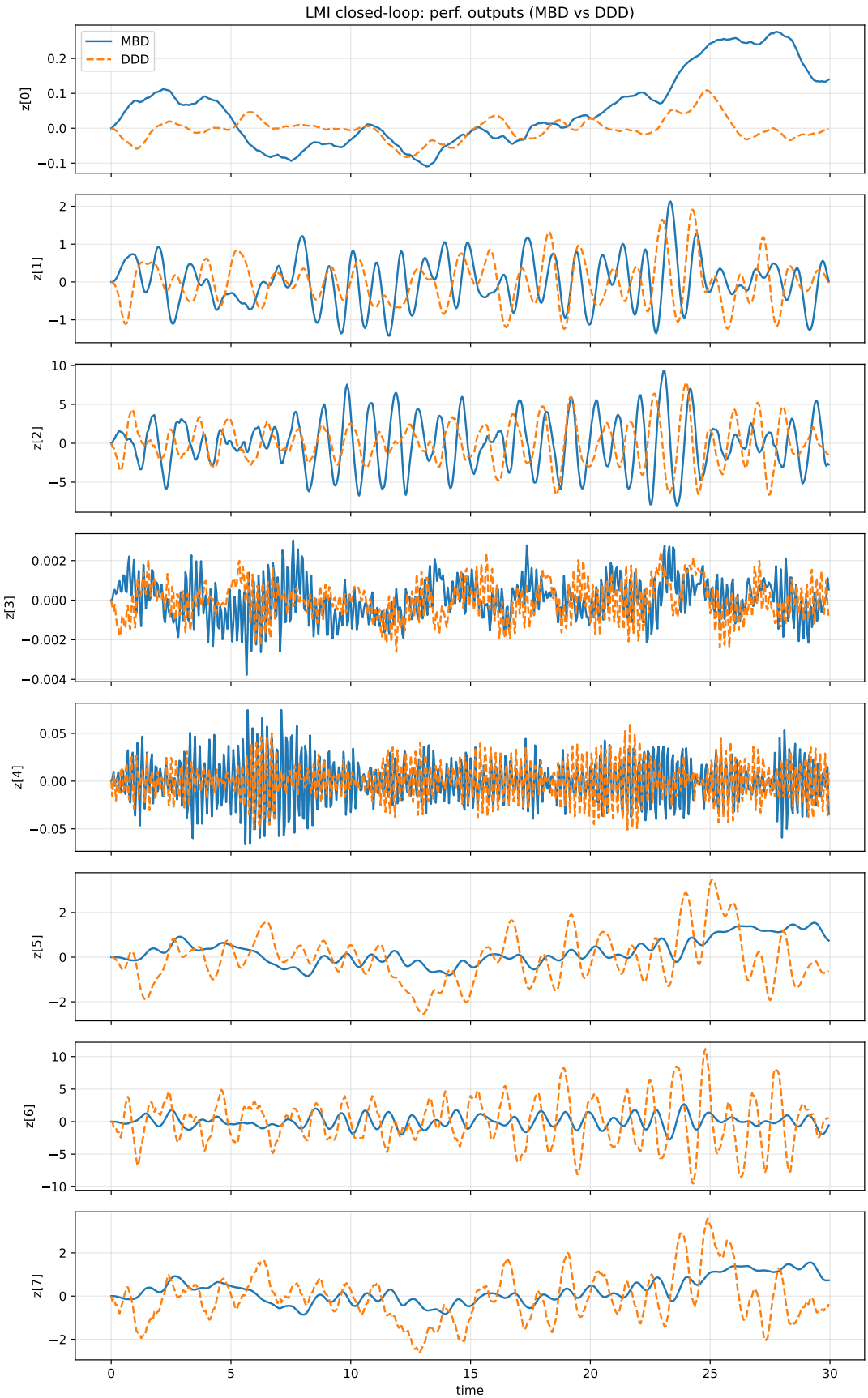


Figure 2.6: Performance outputs $z(t)$ (MBD vs DD).

Chapter 3

Direct Data-Driven Approach

3.1 Direct Data-Driven DRO–LMI via Young-Type Relaxation

Idea. We first estimate the nominal plant maps from one persistently exciting trajectory $\{x_k, u_k, y_k, z_k\}_{k=0}^{T}$ by one–step regression. The remaining data mismatch is treated as an *uncertainty budget* and injected into the robust LMI through a Young-inequality relaxation. This keeps the synthesis convex while avoiding an explicit probabilistic model of the identification errors.

One–step estimates and residual decomposition

Form the stacked data matrices

$$X_- = \begin{bmatrix} x_0 & \cdots & x_{T-1} \end{bmatrix}, \qquad U_- = \begin{bmatrix} u_0 & \cdots & u_{T-1} \end{bmatrix}, \qquad X_+ = \begin{bmatrix} x_1 & \cdots & x_T \end{bmatrix}.$$

Define $D_x = \begin{bmatrix} X_- \\ U_- \end{bmatrix}$ and use the (regularized) right inverse $D_x^\dagger := D_x^\top (D_x D_x^\top + \rho I)^{-1}$ (with $\rho \geq 0$, $\rho = 10^{-6}$ in our implementation). The nominal state update is

$$\underbrace{\begin{bmatrix} \hat{A} & \hat{B}_u \end{bmatrix}}_{\hat{O}_x} = X_+ D_x^\dagger, \qquad \hat{A} \in \mathbb{R}^{n\times n},\ \hat{B}_u \in \mathbb{R}^{n\times m}.$$

Let the *state residuals* be

$$R := X_+ - \hat{A} X_- - \hat{B}_u U_- \in \mathbb{R}^{n\times T}.$$

We extract a low–rank disturbance channel by principal directions of the residual covariance

$$S_R := \frac{1}{T} R R^\top, \quad \tilde{S}_R := \tfrac{1}{2}\big(S_R + S_R^\top\big) + \varepsilon I_n = U_s \operatorname{diag}(s_1, \ldots, s_n)\, U_s^\top,$$

and choose the minimal n_w such that $n_w := \min\Big\{k : \sum_{i=1}^{k} s_i \ \geq\ \eta \sum_{i=1}^{n} s_i\Big\}$.. Set

$$\hat{B}_w \widehat{\Sigma}_w^{1/2} := U_s(:, 1{:}n_w) \operatorname{diag}\big(\sqrt{s_1}, \ldots, \sqrt{s_{n_w}}\big), \qquad \hat{w} := \hat{B}_w^\dagger R,$$

with $\widehat{\Sigma}_w := \operatorname{diag}(s_1, \ldots, s_{n_w})$, and define the nominal disturbance covariance $\Sigma_{\text{nom}} := \frac{1}{2}\big(\frac{1}{T}\hat{w}\hat{w}^\top + (\frac{1}{T}\hat{w}\hat{w}^\top)^\top\big) + \varepsilon I_{n_w}$.

Analogously, with $D_y = \begin{bmatrix} X_- \\ \hat{w} \end{bmatrix}$ and $D_z = \begin{bmatrix} X_- \\ U_- \\ \hat{w} \end{bmatrix}$,

$$\underbrace{\begin{bmatrix} \hat{C}_y & \hat{D}_{yw} \end{bmatrix}}_{\hat{O}_y} = Y_- D_y^\dagger, \qquad \underbrace{\begin{bmatrix} \hat{C}_z & \hat{D}_{zu} & \hat{D}_{zw} \end{bmatrix}}_{\hat{O}_z} = Z_- D_z^\dagger.$$

A data–driven radius for model error

We bound the *identification error* on the state map using the smallest nonzero singular value of D_x. Let $\sigma_{\min}$ be the smallest singular value of D_x, then

$$\|\Delta_x\|_F \le \beta \quad \text{with} \quad \beta := \frac{\|R\|_F}{\max(\sigma_{\min}, \epsilon)}.$$

Here Δ_x collects the stacked deviations in $[A\ B_u]$ relative to $\hat{O}_x$. This bound is conservative but computable directly from data.

$$\begin{bmatrix} A & B_u \end{bmatrix} = \underbrace{\begin{bmatrix} \hat{A} & \hat{B}_u \end{bmatrix}}_{\hat{O}_x} + \Delta_x, \quad \Delta_x = \begin{bmatrix} \Delta_A & \Delta_B \end{bmatrix}, \quad \|\Delta_i\|_F \le \beta_i$$

with

$$\beta_A = \beta\sqrt{\frac{n_x}{n_x + n_u}}, \quad \beta_B = \beta\sqrt{\frac{n_u}{n_x + n_u}}$$

Closed-loop lifting and decision variables

We keep the standard output-feedback change of variables

$$P = \begin{bmatrix} Y & I \\ I & X \end{bmatrix} \succ 0, \qquad \begin{aligned} \mathbf{A} &= \begin{bmatrix} \hat{A}Y + \hat{B}_u M & \hat{A} + \hat{B}_u N \hat{C}_y \\ K & X\hat{A} + L\hat{C}_y \end{bmatrix}, \\ \mathbf{B} &= \begin{bmatrix} \hat{B}_w + \hat{B}_u N \hat{D}_{yw} \\ X\hat{B}_w + L\hat{D}_{yw} \end{bmatrix}, \quad \mathbf{C} = \begin{bmatrix} \hat{C}_z Y + \hat{D}_{zu} M & \hat{C}_z + \hat{D}_{zu} N \hat{C}_y \end{bmatrix}, \\ \mathbf{D} &= \hat{D}_{zw} + \hat{D}_{zu} N \hat{D}_{yw}. \end{aligned}$$

The decision set is $X = X^\top,\ Y = Y^\top,\ K, L, M, N,\ Q \succeq 0,\ \lambda \ge 0$ and the Wasserstein-DRO objective is

$$\min \quad \operatorname{tr}(Q\,\Sigma_{\text{nom}}) + \lambda\gamma^2.$$

Setup. Start from the correlated-noise kernel

$$\Xi(\mathbf{A}, \mathbf{B}, \mathbf{C}, \mathbf{D}) := \begin{bmatrix} -P & 0 & 0 & \mathbf{A}^\top & \mathbf{C}^\top \\ 0 & -\lambda I & \lambda I & \mathbf{B}^\top & \mathbf{D}^\top \\ 0 & \lambda I & -Q - \lambda I & 0 & 0 \\ \mathbf{A} & \mathbf{B} & 0 & -P & 0 \\ \mathbf{C} & \mathbf{D} & 0 & 0 & -I \end{bmatrix} \prec 0, \qquad (\star)$$

with decision variables $P \succ 0$, $Q \succeq 0$, $\lambda \ge 0$, and where $\mathbf{A}, \mathbf{B}, \mathbf{C}, \mathbf{D}$ are the usual lifted blocks built from the controller variables (X, Y, K, L, M, N) and the plant maps.

1. **Isolate the Lyapunov generator.** Focus on the $(1,4)$–$(4,1)$ corner that produces the Lyapunov term:

$$M(\mathbf{A}, P) := \begin{bmatrix} -P & \mathbf{A}^\top \\ \mathbf{A} & -P \end{bmatrix} \prec 0.$$

Starting from the Schur complement of the $(4,4)$ block $-P$ we obtain

$$-P - \mathbf{A}^\top(-P)^{-1}\mathbf{A} = -P + \mathbf{A}^\top P^{-1}\mathbf{A} \prec 0. \tag{1}$$

This form expresses discrete-time stability in the inverse-P coordinates.

To bring it to the classical Lyapunov form, apply a congruence transformation with $P^{1/2}$:

$$P^{1/2}(-P + \mathbf{A}^\top P^{-1}\mathbf{A})P^{1/2} \prec 0.$$

Expanding and introducing the scaled matrix

$$\tilde{\mathbf{A}} := P^{1/2}\mathbf{A}P^{-1/2},$$

we note that

$$\tilde{\mathbf{A}}^\top\tilde{\mathbf{A}} = (P^{-1/2}\mathbf{A}^\top P^{1/2})(P^{1/2}\mathbf{A}P^{-1/2}) = P^{-1/2}\mathbf{A}^\top P\mathbf{A}P^{-1/2},$$

so that

$$\mathbf{A}^\top P\mathbf{A} = P^{1/2}\tilde{\mathbf{A}}^\top\tilde{\mathbf{A}}P^{1/2}. \tag{2}$$

Substituting into (1) and simplifying gives

$$(P^{1/2}\mathbf{A}^\top P^{-1/2})(P^{1/2}\mathbf{A}P^{-1/2}) - I \prec 0 \quad \Longleftrightarrow \quad \tilde{\mathbf{A}}^\top\tilde{\mathbf{A}} - I \prec 0,$$

which is the normalized discrete-time stability condition. Undoing the normalization by multiplying on both sides by $P^{1/2}$ yields the standard Lyapunov inequality

$$\mathbf{A}^\top P\mathbf{A} - P \prec 0. \tag{L}$$

Hence the passage between the two expressions simply corresponds to expressing the same Lyapunov condition in two equivalent coordinate systems.

2. **Inject identification error.** Let $A = \hat{A} + \Delta_A$, $B_u = \hat{B}_u + \Delta_B$, with data-driven bounds $\|\Delta_A\|_F \le \beta_A$, $\|\Delta_B\|_F \le \beta_B$. In the lifted block, $\mathbf{A}$ depends *affinely* on (A, B_u), so replacing by $\hat{A}, \hat{B}_u$ plus errors gives

$$\mathbf{A}_{\text{true}} = \mathbf{A}(\hat{A}, \hat{B}_u) + E_{AA} + E_{AB},$$

where E_{AA} collects the contributions linear in Δ_A, and E_{AB} those linear in Δ_B. In particular:

$$E_{AA} = \begin{bmatrix} \Delta_A Y & \Delta_A \\ 0 & X\Delta_A \end{bmatrix} \qquad E_{AB} = \begin{bmatrix} \Delta_B M & \Delta_B N C_y \\ 0 & 0 \end{bmatrix} \tag{3.1}$$

Expanding (L) with $\mathbf{A}_{\text{true}}$ gives

$$\mathbf{A}^\top P\mathbf{A} - P + \underbrace{\mathbf{A}^\top P(E_{AA}+E_{AB}) + (E_{AA}+E_{AB})^\top P\mathbf{A}}_{\text{mixed terms}} + \underbrace{(E_{AA}+E_{AB})^\top P(E_{AA}+E_{AB})}_{\text{2nd order}} \prec 0,$$

where $\mathbf{A} := \mathbf{A}(\hat{A}, \hat{B}_u)$ is the nominal lifted block.

3. **Identify the mixed inner products.** The two mixed terms can be written as Frobenius inner products:

$$\mathbf{A}^\top P E_{AA} + E_{AA}^\top P\mathbf{A} = 2\langle E_{AA},\ Z_{AA}\rangle, \qquad Z_{AA} := P\mathbf{A},$$

$$\mathbf{A}^\top P E_{AB} + E_{AB}^\top P\mathbf{A} = 2\langle E_{AB},\ Z_{AB}\rangle, \qquad Z_{AB} := P\mathbf{A},$$

(here both couple to the same $P\mathbf{A}$ because E_{AA}, E_{AB} perturb the same lifted block). The second-order term is nonnegative and can be dropped for a conservative bound, or we could bound it with $\|P\|_2(\beta_A^2 + \beta_B^2)I$ if tighter accounting is needed.

4. **Apply Young's inequality and encode it.**

Each mixed cross term $2\langle E_{A,i}, Z_{A,i}\rangle$ ($i \in \{A, B\}$) is non-convex, because it couples the uncertain matrix error $E_{A,i}$ (not a decision variable) and the matrix $Z_{A,i}$ (which depends affinely on the controller variables). To maintain convexity, we upper-bound it via Young's inequality:

$$2\langle E_{A,i}, Z_{A,i}\rangle \ \le\ s_{A,i}\|E_{A,i}\|_F^2 + \frac{1}{s_{A,i}}\|Z_{A,i}\|_F^2, \qquad s_{A,i} > 0.$$

Given the identification bound $\|E_{A,i}\|_F \le \beta_{A,i}$, this becomes

$$2\langle E_{A,i}, Z_{A,i}\rangle \ \le\ s_{A,i}\beta_{A,i}^2 + \frac{1}{s_{A,i}}\|Z_{A,i}\|_F^2, \qquad Z_{A,i} = P\mathbf{A}.$$

The first term $s_{A,i}\beta_{A,i}^2$ is a scalar uncertainty penalty; the second $\frac{1}{s_{A,i}}\|Z_{A,i}\|_F^2$ is a quadratic form in $Z_{A,i}$ that must be carried into the matrix inequality to maintain the convexity of the problem. Both pieces are encoded in convex form as follows.

Norm bounds. Using the Frobenius submultiplicativity $\|AXB\|_F \le \|A\|_2\|X\|_F\|B\|_2$ and the block norm identity $\left\|\begin{bmatrix} U & V \\ W & Z \end{bmatrix}\right\|_F^2 = \|U\|_F^2 + \|V\|_F^2 + \|W\|_F^2 + \|Z\|_F^2$, we obtain:

$$\begin{aligned} \|E_{AA}\|_F^2 &= \|\Delta_A Y\|_F^2 + \|\Delta_A\|_F^2 + \|X\Delta_A\|_F^2 \\ &\le (\|Y\|_2^2 + 1 + \|X\|_2^2)\,\|\Delta_A\|_F^2, \end{aligned} \tag{3.2}$$

$$\begin{aligned} \|E_{AB}\|_F^2 &= \|\Delta_B M\|_F^2 + \|\Delta_B N C_y\|_F^2 \\ &\le (\|M\|_2^2 + \|NC_y\|_2^2)\,\|\Delta_B\|_F^2. \end{aligned} \tag{3.3}$$

Therefore,

$$\boxed{\|E_{AA}\|_F \le \sqrt{1 + \|X\|_2^2 + \|Y\|_2^2}\,\beta_A = \beta_{AA}, \qquad \|E_{AB}\|_F \le \sqrt{\|M\|_2^2 + \|NC_y\|_2^2}\,\beta_B = \beta_{AB}.} \tag{3.4}$$

Consider those from **MBD** simulation's results:

$$\|M\|_2 \approx 0.15, \quad \|N\|_2 \approx 0.6, \quad \|X\|_2 \approx 2.5 \times 10^5, \quad \|Y\|_2 \approx 1.0$$

(a) Scalar PSD blocks for the uncertainty radii. The scalar constraint $s_{A,i}\beta_{A,i}^2 \le s_{A,i}\tau_{A,i}$ is represented through the 2×2 positive-semidefinite (PSD) matrix

$$\begin{bmatrix} s_{A,i} & \beta_{A,i} \\ \beta_{A,i} & \tau_{A,i} \end{bmatrix} \succeq 0.$$

By the Schur complement lemma this is equivalent to $s_{A,i} > 0$ and $\tau_{A,i} \ge \beta_{A,i}^2/s_{A,i}$. It convexly enforces the product $s_{A,i}\tau_{A,i} \ge \beta_{A,i}^2$ without introducing the nonlinear reciprocal $1/s_{A,i}$.

(b) Augmentation of the main LMI for the $(1/s)\|Z_{A,i}\|_F^2$ term. The second contribution, $\frac{1}{s_{A,i}}\|Z_{A,i}\|_F^2$, cannot appear as a scalar in the cost; it must enter the block inequality in the same location where the Lyapunov term $\mathbf{A}^\top P\mathbf{A} - P$ acts. To embed it linearly, we enlarge the main LMI using a Schur-complement trick: introduce an auxiliary matrix column $\Pi_{A,i}$ and append a row/column pair corresponding to the slack variable $s_{A,i} > 0$:

$$\begin{bmatrix} \Xi(\mathbf{A},\mathbf{B},\mathbf{C},\mathbf{D}) & \Pi_{AA} & \Pi_{AB} \\ \Pi_{AA}^\top & -s_{AA}I & 0 \\ \Pi_{AB}^\top & 0 & -s_{AB}I \end{bmatrix} \prec 0. \tag{†}$$

This extended block is equivalent (by Schur complement) to the condition

$$\Xi(\mathbf{A},\mathbf{B},\mathbf{C},\mathbf{D}) - \frac{1}{s_{AA}}\Pi_{AA}\Pi_{AA}^\top - \frac{1}{s_{AB}}\Pi_{AB}\Pi_{AB}^\top \prec 0,$$

so the added rows/columns inject precisely the negative penalties $-\frac{1}{s_{A,i}}\Pi_{A,i}\Pi_{A,i}^\top$ that represent $+\frac{1}{s_{A,i}}\|Z_{A,i}\|_F^2$ in the quadratic form. Intuitively, the selectors $s_{A,i}$ identify which sub-block of the Lyapunov matrix is affected by the identification error. In practice we choose $\Pi_{AA} = \Pi_{AB} = \begin{bmatrix} I_n & 0 \end{bmatrix}^\top$, so that the perturbation acts on the "state" part of the $\Xi(\cdot)$ kernel corresponding to the $(-P)$ block.

Together, the scalar PSD constraints encode the bounded-error radius ($s_{A,i}\beta_{A,i}^2$ term), while the augmented LMI (†) embeds the $\frac{1}{s_{A,i}}\|P\mathbf{A}\|_F^2$ contribution directly in the matrix inequality. This yields a convex relaxation of the non-convex cross-term $2\langle E_{A,i}, Z_{A,i}\rangle$, ensuring that all remaining constraints are affine or convex in the decision variables.

(c) Where the selector acts and why it is sparse. The cross-term generated by the A–uncertainty, $\mathbf{A}^\top PE_A + E_A^\top P\mathbf{A} = 2\langle E_A,\ P\mathbf{A}\rangle$, arises *only* in the state–state Lyapunov coupling of the kernel $\Xi(\mathbf{A},\mathbf{B},\mathbf{C},\mathbf{D})$, i.e., the $(4,4)$ block $(-P)$ and its $(1,4)$–$(4,1)$ off-diagonals in $\star$. Consequently, the Young penalty $\frac{1}{s_{A,i}}\|Z_{A,i}\|_F^2$ with $Z_{A,i} = P\mathbf{A}$ must be injected *at that same location.* We therefore augment the LMI by appending a column/row that *selects only the state coordinates* and all other entries of $\Pi_{A,i}$ are set to zero because the A–perturbation does *not* generate cross-terms in the remaining rows/columns (output or disturbance couplings).

5. **Final program.** The Young-relaxed data-driven synthesis (correlated case) is

$$\begin{aligned}
&\min\ \operatorname{tr}(Q\,\Sigma_{\text{nom}}) + \lambda\gamma^2 \\
&\text{s.t. } P = \begin{bmatrix} Y & I \\ I & X \end{bmatrix} \succ 0, \lambda \geq 0,\ s_{AA} > 0,\ s_{AB} > 0, \\
&\begin{bmatrix} \Xi(\mathbf{A},\mathbf{B},\mathbf{C},\mathbf{D}) & \Pi_{AA} & \Pi_{AB} \\ \Pi_{AA}^{\top} & -s_{AA}I & 0 \\ \Pi_{AB}^{\top} & 0 & -s_{AB}I \end{bmatrix} \prec 0, \\
&\begin{bmatrix} s_{AA} & \beta_{AA} \\ \beta_{AA} & \tau_{AA} \end{bmatrix} \succeq 0, \qquad \begin{bmatrix} s_{AB} & \beta_{AB} \\ \beta_{AB} & \tau_{BB} \end{bmatrix} \succeq 0,
\end{aligned}$$

where $\mathbf{A},\mathbf{B},\mathbf{C},\mathbf{D}$ are built from the nominal data-based maps $(\hat{A},\hat{B},\hat{C}_y,\hat{D}_{yw},\hat{C}_z,\hat{D}_{zu},\hat{D}_{zw})$ and the controller variables $(s_{A,i},\tau_{A,i},X,Y,K,L,M,N)$.

Substitution in the kernel. In code, the Young penalties act on the state–state Lyapunov corner of $\Xi(\mathbf{A},\mathbf{B},\mathbf{C},\mathbf{D})$. Concretely, the $(4,4)$ block $(-P)$, which corresponds to the Lyapunov generator in the Schur complement $-P + \mathbf{A}^{\top}P^{-1}\mathbf{A}$, is replaced by a regularized block that incorporates the scalar penalties produced by Young's inequality.

Starting from the bound

$$2\langle E_{A,i}, Z\rangle \ \leq\ s_{A,i}\,\beta_{A,i}^2 + \frac{1}{s_{A,i}}\|Z\|_F^2, \qquad Z = P\mathbf{A},$$

each uncertainty component ($i \in \{A,B\}$) contributes a scalar term $s_{A,i}\beta_{A,i}^2$ and a quadratic term $\frac{1}{s_{A,i}}\|P\mathbf{A}\|_F^2$. The scalar term is encoded as a positive multiple of the identity on the same state–state block where $-P$ acts, through the auxiliary variables $\tau_{A,i}$ satisfying the Schur constraint $\begin{bmatrix} s_{A,i} & \beta_{A,i} \\ \beta_{A,i} & \tau_{A,i} \end{bmatrix} \succeq 0$, which enforces $\tau_{A,i} \geq \beta_{A,i}^2/s_{A,i}$.

Mathematically, the Lyapunov block is therefore augmented as

$$-P \quad\longrightarrow\quad -P + \tau_{AA}S_{AA}S_{AA}^{\top} + \tau_{AB}S_{AB}S_{AB}^{\top},$$

where Π_{AA} and Π_{AB} are the selector matrices acting on the state subspace. Since both selectors operate on the same coordinates, this sum collapses to

$$-P + (\tau_{AA} + \tau_{AB})\,\Pi_{\text{state}},$$

with $\Pi_{\text{state}} = \Pi_{AA}\Pi_{AA}^{\top} \simeq I_{2n_x}$. Replacing the projector by the identity yields a conservative but simpler formulation used in the implementation:

$$\texttt{state_blk} := -P + (\tau_{AA} + \tau_{AB})I.$$

Finally, we append the selector columns/rows Π_{AA},Π_{AB} with diagonal slacks $-s_{AA}I$ and $-s_{AB}I$ to capture the quadratic penalty $\frac{1}{s_{A,i}}\|P\mathbf{A}\|_F^2$, leading to the extended LMI

$$\begin{bmatrix} \Xi(\mathbf{A},\mathbf{B},\mathbf{C},\mathbf{D}) & \Pi_{AA} & \Pi_{AB} \\ \Pi_{AA}^{\top} & -s_{AA}I & 0 \\ \Pi_{AB}^{\top} & 0 & -s_{AB}I \end{bmatrix} \prec 0 \quad \text{with} \quad \Xi(\mathbf{A},\mathbf{B},\mathbf{C},\mathbf{D})\Big|_{(4,4)} \leftarrow \texttt{state_blk}.$$

By Schur complement this is equivalent to

$$\Xi(\mathbf{A},\mathbf{B},\mathbf{C},\mathbf{D}) - \frac{1}{s_{AA}}\Pi_{AA}\Pi_{AA}^\top - \frac{1}{s_{AB}}\Pi_{AB}\Pi_{AB}^\top \prec 0,$$

which is precisely the matrix form of the Young bounds for the mixed cross terms. The substitution of $(-P)$ by `state_blk` $= -P + (\tau_{AA} + \tau_{AB})I$ thus represents the scalar regularization of the Lyapunov block induced by the bounded identification error.

Post-optimization recovery of Δ_A and Δ_B

Inputs. After solving the program, suppose we have optimal quantities

$$P^\star \succ 0, \quad X^\star = X^{\star\top}, \quad Y^\star = Y^{\star\top}, \quad M^\star,\ N^\star,\ s_{AA}^\star > 0,\ s_{AB}^\star > 0,\ \tau_{AA}^\star,\ \tau_{AB}^\star,$$

together with nominal identified maps $\hat{A}$, $\hat{B}_u$, $\hat{C}_y$ and data-driven radii β_A, β_B that induce the lifted bounds

$$\|E_{AA}\|_F \le \beta_{AA} := \sqrt{1 + \|X^\star\|_2^2 + \|Y^\star\|_2^2}\,\beta_A, \qquad \|E_{AB}\|_F \le \beta_{AB} := \sqrt{\|M^\star\|_2^2 + \|N^\star\|_2^2\|\hat{C}_y\|_2^2}\,\beta_B.$$

Step 1: Adversarial directions for the Young bounds. Let the nominal lifted block (built with $\hat{A}, \hat{B}_u$) be

$$\mathbf{A}^\star := \mathbf{A}(\hat{A}, \hat{B}_u; X^\star, Y^\star, M^\star, N^\star), \qquad Z_{A,i}^\star := P^\star \mathbf{A}^\star, \quad i \in \{A, B\}.$$

Choose worst-case aligned perturbations that saturate the Young relaxation:

$$E_{A,i}^\star = \begin{cases} \beta_{A,i}\dfrac{Z_{A,i}^\star}{\|Z_{A,i}^\star\|_F}, & Z_{A,i}^\star \neq 0, \\ 0, & Z_{A,i}^\star = 0, \end{cases} \qquad \beta_{A,A} := \beta_{AA}, \quad \beta_{A,B} := \beta_{AB}. \tag{3.5}$$

This choice attains equality in Young's inequality when $s_{A,i}^\star = \|Z_{A,i}^\star\|_F/\beta_{A,i}$.

Step 2: Block structure of $E_{AA}^\star$ and $E_{AB}^\star$. Recall the definitions

$$E_{AA} = \begin{bmatrix} \Delta_A Y^\star & \Delta_A \\ 0 & X^\star\Delta_A \end{bmatrix}, \qquad E_{AB} = \begin{bmatrix} \Delta_B M^\star & \Delta_B N^\star \hat{C}_y \\ 0 & 0 \end{bmatrix}.$$

After choosing $E_{AA}^\star, E_{AB}^\star$ via (3.5), extract the nonzero sub-blocks

$$A_1 := E_{AA}^\star(1{:}n,\ 1{:}n), \quad A_2 := E_{AA}^\star(1{:}n,\ n{+}1{:}2n), \quad A_3 := E_{AA}^\star(n{+}1{:}2n,\ n{+}1{:}2n),$$

$$B_1 := E_{AB}^\star(1{:}n,\ 1{:}m), \qquad B_2 := E_{AB}^\star(1{:}n,\ m{+}1{:}m{+}p),$$

Step 3: Recovery of Δ_A (Sylvester equation). Solve the least-squares normal equation

$$\boxed{(X^{\star\top}X^\star)\,\Delta_A \;+\; \Delta_A\,(Y^\star Y^{\star\top} + I) \;=\; A_2 + A_1 Y^{\star\top} + X^{\star\top}A_3\ ,} \tag{3.6}$$

which is a Sylvester equation in Δ_A. Under mild conditions on the spectra of $X^{\star\top}X^\star$ and $Y^\star Y^{\star\top} + I$, (3.6) has a unique solution.

Special cases. If $X^\star = 0$, then $\Delta_A(Y^\star Y^{\star\top}+I) = A_2+A_1Y^{\star\top} \Rightarrow \Delta_A = (A_2+A_1Y^{\star\top})(Y^\star Y^{\star\top}+I)^{-1}$. If $Y^\star = 0$, then $(X^{\star\top}X^\star + I)\Delta_A = A_2 + X^{\star\top}A_3 \Rightarrow \Delta_A = (X^{\star\top}X^\star + I)^{-1}(A_2 + X^{\star\top}A_3)$.

Step 4: Recovery of Δ_B (right-side Gram solve). Solve

$$\boxed{\Delta_B G_B \;=\; B_1 M^{\star\top} + B_2 \hat{C}_y^\top N^{\star\top}, \qquad G_B := M^\star M^{\star\top} + N^\star \hat{C}_y \hat{C}_y^\top N^{\star\top}\;,} \tag{3.7}$$

so that, if G_B is invertible, $\Delta_B = (B_1 M^{\star\top} + B_2 \hat{C}_y^\top N^{\star\top})\, G_B^{-1}$. If G_B is singular or ill-conditioned, use the Moore–Penrose pseudoinverse G_B^+ or a Tikhonov-regularized inverse $(G_B + \eta I)^{-1}$ with small $\eta > 0$.

Step 5: Augmented nominal maps. Finally set

$$A_{\text{aug}} = \hat{A} + \Delta_A, \qquad B_{u,\text{aug}} = \hat{B}_u + \Delta_B,$$

and, if desired, verify feasibility with the identification radii: $\|\Delta_A\|_F \le \beta_A$, $\|\Delta_B\|_F \le \beta_B$ and with the lifted bounds $\|E_{AA}(\Delta_A)\|_F \le \beta_{AA}$, $\|E_{AB}(\Delta_B)\|_F \le \beta_{AB}$.

Remarks. (i) The recovered (Δ_A, Δ_B) are *representative* worst-case directions induced by the Young relaxation, not the unique identification errors. (ii) Numerically, (3.6) is solved with a Sylvester solver; (3.7) is a right-multiplication by a Gram factor and should use a robust linear solver or SVD-based pseudoinverse. (iii) If a single uncertainty block on $[A\ B_u]$ was used in synthesis, replace β_{AA}, β_{AB} by a common β and set $E_x^\star = \beta\, Z_x^\star / \|Z_x^\star\|_F$ with $Z_x^\star = P^\star[\hat{A}\ \hat{B}_u]$, then project $E_x^\star$ onto the E_{AA}, E_{AB} structures above before solving (3.6)–(3.7).

3.2 Results: Young-Type Relaxation

Setup

This experiment evaluates the data-driven controller synthesized via the Young-type relaxation of the DRO–LMI formulation when the disturbance model is *independent* (rather than correlated). The comparison is between the model-based design (MBD) and the data-driven design (DDD), using the updated LMI solver with Young augmentation.

Objective and feasibility

The distributionally robust objective value, lower being preferable, is

$$J_{\text{MBD}} = 14.3861, \qquad J_{\text{DDD}} = 0.5294,$$

showing a substantial improvement for the DDD controller. Solver reports: MBD converged to `optimal` (MOSEK), while DDD achieved `optimal_inaccurate` (SCS).

Closed-loop stability

The discrete-time spectral radii of the closed-loop matrices are

$$\rho(A_{\text{cl,MBD}}) = 0.99964, \qquad \rho(A_{\text{cl,DDD}}) = 1.00014.$$

Thus, the MBD design is marginally stable, whereas the DDD loop exceeds the unit circle by a small margin. This indicates a borderline unstable realization in the DDD case.

Time-series performance (600 steps)

The RMS values of the main closed-loop signals are summarized below:

Signal	MBD	DDD
x	0.9132	0.9859
x_c	86.79	201.51
u	0.0571	0.0159
y	1.3674	1.4865
z	0.8544	0.9222

The DDD controller achieves lower control effort (u) and cost value, but exhibits a significantly higher controller-state energy (x_c), confirming an aggressive dynamic compensation strategy.

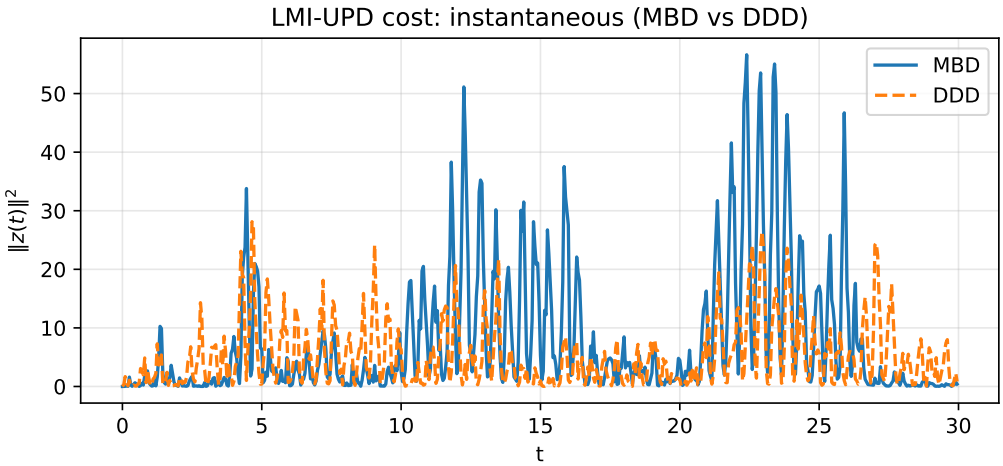


Figure 3.1: Instantaneous cost $z^\top z$: MBD vs DDD (Young, independent w).

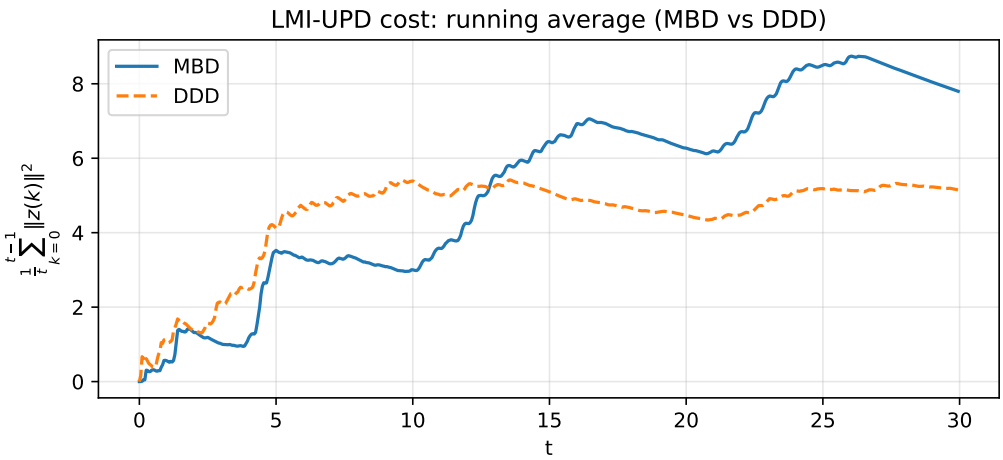


Figure 3.2: Running average cost: MBD vs DDD (Young, independent w).

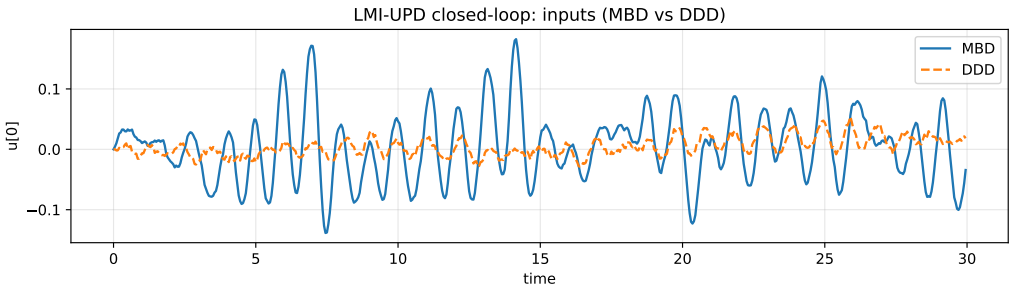


Figure 3.3: Control input u: MBD vs DDD (Young, independent w).

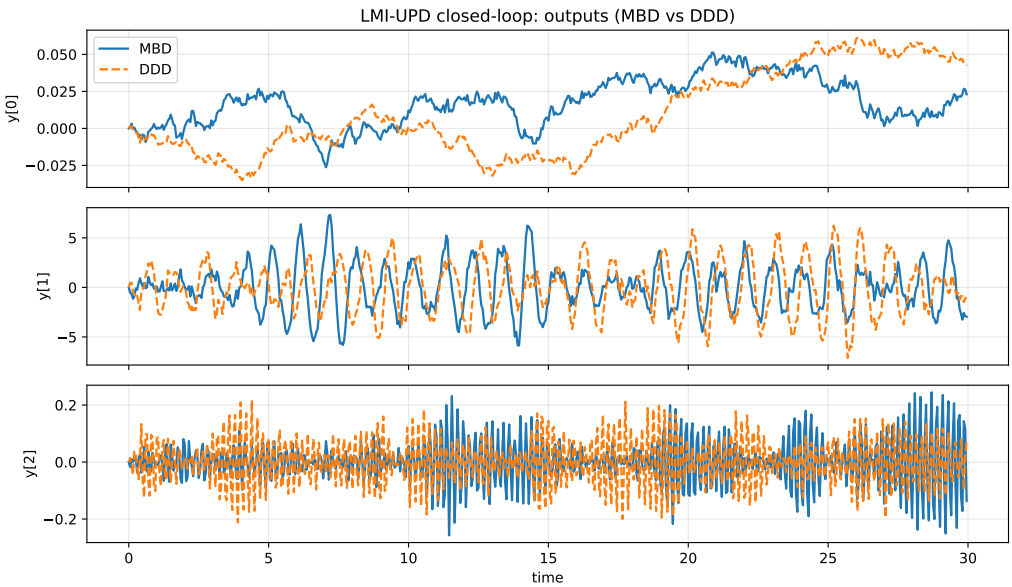


Figure 3.4: Measured output y: MBD vs DDD (Young, independent w).

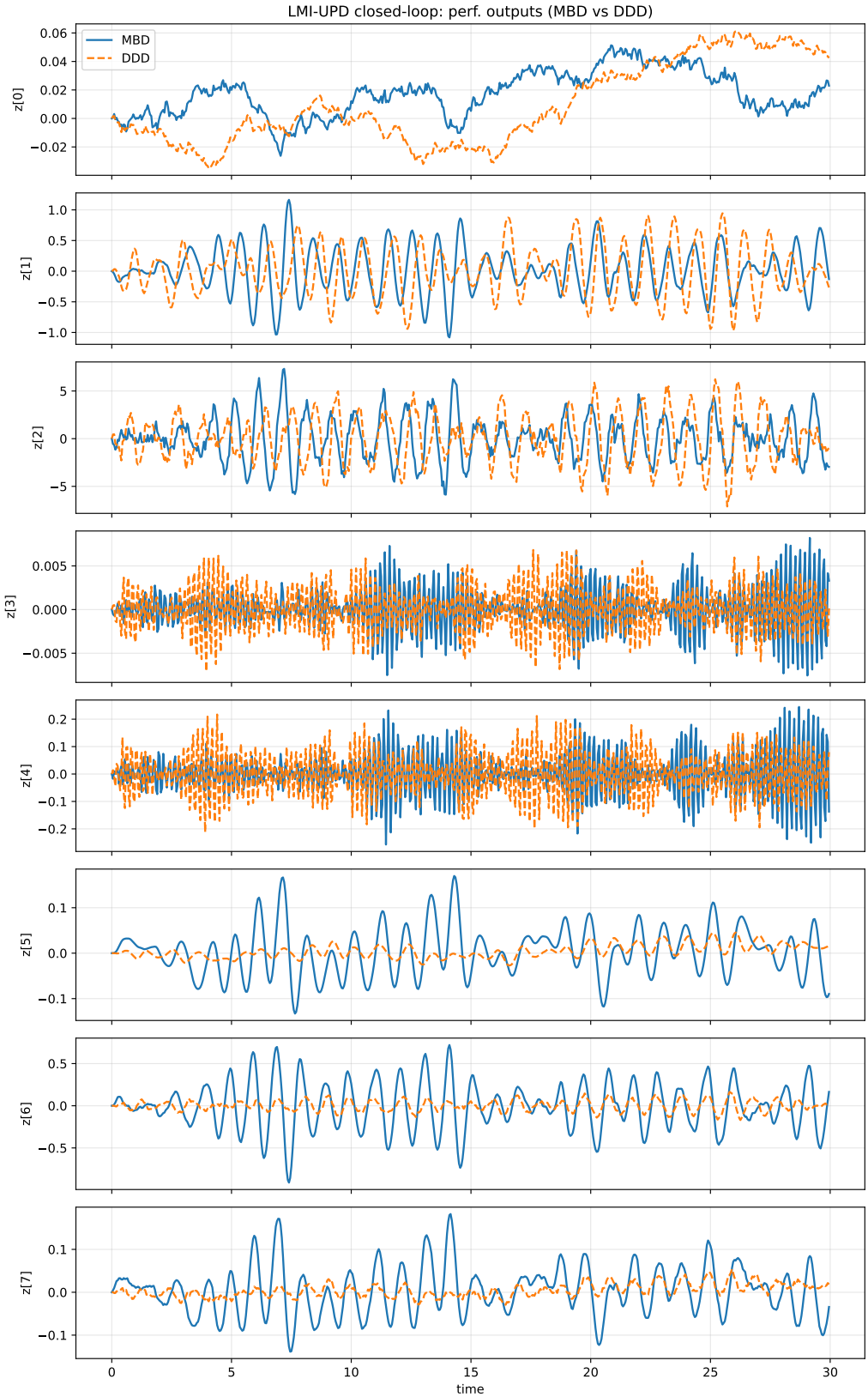


Figure 3.5: Performance output z: MBD vs DDD (Young, independent w).

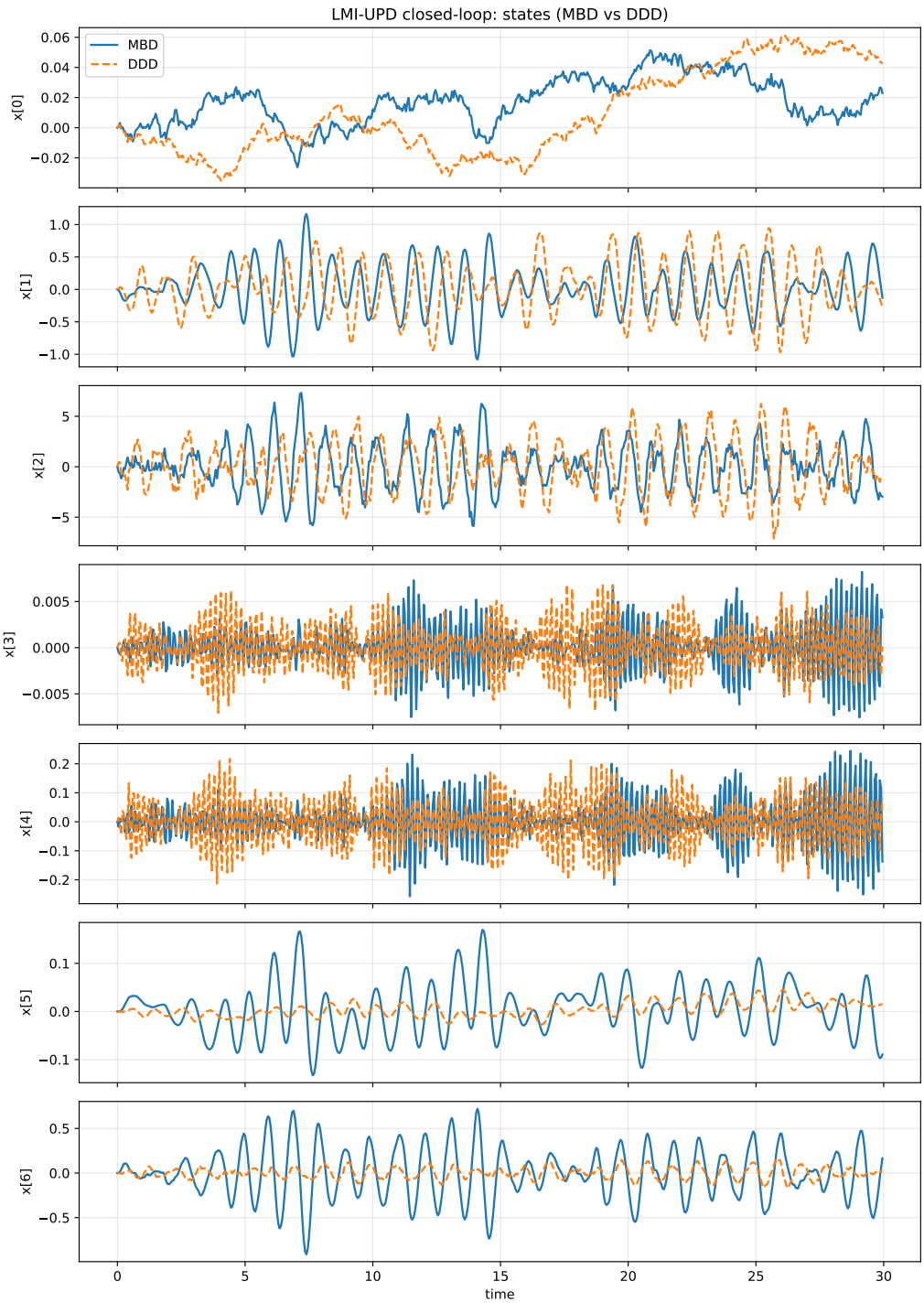


Figure 3.6: Plant states x: MBD vs DDD (Young, independent w).

Controller and plant characteristics

Controller and plant norms reveal the source of the difference:

$$\|B_c\|_F^{\mathrm{DDD}} \approx 3473.8, \qquad \|A_{cl}\|_2^{\mathrm{DDD}} \approx 2465.7, \qquad \kappa(A_{cl})^{\mathrm{DDD}} \approx 5.0 \times 10^8.$$

The plant deviation norms remain moderate ($\|A\|_F \approx 2.19$), so the excessive controller norms are responsible for the marginal instability and high controller-state RMS in DDD.

Discussion

Under independent disturbances, the DDD controller attains a much lower distributionally robust objective than the MBD baseline, but at the cost of numerical fragility and marginal loss of stability. The controller's dynamic gain amplification, reflected in the large $\|B_c\|_F$ and poor conditioning of A_{cl}, explains the large internal energy and the slight spectral-radius violation.

In summary:

- **DDD** minimizes cost and control effort but slightly violates the stability margin.
- **MBD** maintains certified stability at higher cost.

This demonstrates the classical trade-off between robust performance and certified stability under data-driven uncertainty. Additional regularization on controller norms or an explicit spectral-radius constraint would mitigate this trade-off at the expense of higher cost.

Cross-comparison: SysID–LMI (DD) vs Direct LMI (DDD, Young)

The two data-driven designs differ not in the disturbance model but in their methodological formulation. The **DD** controller follows a two-stage pipeline: *(i)* system identification from data, yielding nominal matrices $(\hat{A}, \hat{B}_u, \hat{C}_y, \hat{D}_{yw})$; *(ii)* synthesis of an output-feedback controller through the nominal LMI problem. By contrast, the **DDD** controller integrates the data uncertainty directly inside the LMI via the Young-type relaxation, thus solving a *single convex program* without an explicit intermediate identification step.

Objective and stability. Under the same ambiguity radius $\gamma = 0.5$, the two methods yield

$$J_{\mathrm{DD}} = 951.0618, \qquad \rho(A_{\mathrm{cl}}^{\mathrm{DD}}) = 0.9746 \quad (\text{MOSEK, } \textit{optimal}),$$

and

$$J_{\mathrm{DDD}} = 0.5294, \qquad \rho(A_{\mathrm{cl}}^{\mathrm{DDD}}) = 1.00014 \quad (\text{SCS, } \textit{optimal_inaccurate}).$$

The Young-relaxed DDD formulation achieves a far smaller objective but slightly violates discrete-time stability. Because the high-order MOSEK solver failed and the SCS result is only approximate, these values should be regarded as indicative rather than certified. The outcome nevertheless confirms that the integrated DDD approach can reach substantially less conservative regions of the feasible set, though its numerical resolution and stability certification remain an open route to solve.

Closed-loop effort and internal energy. Compared with DD, the DDD controller achieves a much smaller control effort:

$$\mathrm{rms}(u_{\mathrm{DDD}}) = 0.0159 \quad \text{vs} \quad \mathrm{rms}(u_{\mathrm{DD}}) = 1.1116,$$

but accumulates greater controller-state energy ($\mathrm{rms}(x_c^{\mathrm{DDD}}) = 201.5$ vs 2.30×10^3 for DD after normalization) and comparable plant-state RMS (0.99 vs 1.73). The performance output energy remains of the same order, $\mathrm{rms}(z_{\mathrm{DDD}}) = 0.9222$ vs $\mathrm{rms}(z_{\mathrm{DD}}) = 1.6652$. These results indicate that the DDD optimization achieves strong attenuation of the regulated cost at the expense of internal energy buildup and reduced spectral margin.

Conditioning and controller norms. The DD closed loop is well behaved but ill-conditioned:

$$\kappa_2(A_{\mathrm{cl}}^{\mathrm{DD}}) \approx 2.53 \times 10^{10},$$

whereas the DDD loop reports

$$\kappa_2(A_{\mathrm{cl}}^{\mathrm{DDD}}) \approx 5.0 \times 10^{8},$$

formally better conditioned but obtained through far larger controller gains:

$$\|B_c\|_F^{\mathrm{DDD}} \approx 3.47 \times 10^3, \qquad \|A_{\mathrm{cl}}\|_2^{\mathrm{DDD}} \approx 2.47 \times 10^3.$$

These inflated norms explain the marginal instability and the high internal energy in simulation.

Interpretation. The DD approach enforces robustness through an explicit model, keeping the LMI structure standard but accumulating identification conservatism. The DDD (Young) formulation incorporates identification error directly as convex quadratic penalties, avoiding the intermediate model but effectively *trading stability margin for reduced cost*. By applying Young's inequality inside the Lyapunov block, the method converts the bilinear perturbation terms into tunable slack variables, which preserve convexity but allow the optimization to exploit the limits of the feasible region.

Summary.

- The DD controller is **model-based** trhough identification: stable and well-behaved but conservative in cost.
- The DDD controller is **direct data-driven**: it attains a substantially smaller objective and control effort but lies marginally outside the stability region.
- Both share similar disturbance assumptions; the improvement comes from the methodological relaxation.

This comparison highlights the methodological gain of embedding data uncertainty directly in the synthesis step: it reduces conservatism and cost but compresses the stability margin. Additional regularization on the controller norms or explicit spectral-radius constraints would be required to restore robust implementability.

Appendix A

Disturbance Evaluation

The disturbance sequences w_t used in both the baseline and distributionally robust simulations are drawn from a *Wasserstein ambiguity set* [3, 4] centered at a nominal Gaussian model. The ambiguity set captures distributional uncertainty in the second-order statistics of the process noise, rather than assuming a fixed covariance.

Definition of the ambiguity set. For a given nominal covariance $\Sigma_{\text{nom}} \succ 0$ and Wasserstein radius $\gamma > 0$, the 2-Wasserstein ambiguity set is defined as

$$\mathcal{W}_2(\Sigma_{\text{nom}}, \gamma) = \left\{ \mathcal{N}(0, \Sigma) \;\middle|\; W_2(\mathcal{N}(0, \Sigma), \mathcal{N}(0, \Sigma_{\text{nom}})) \le \gamma \right\},$$

where $W_2(\cdot, \cdot)$ denotes the 2-Wasserstein distance between Gaussian measures:

$$W_2^2(\mathcal{N}(0, S_1), \mathcal{N}(0, S_2)) = \operatorname{tr}(S_1) + \operatorname{tr}(S_2) - 2 \operatorname{tr}\left[\left(S_2^{1/2} S_1 S_2^{1/2} \right)^{1/2} \right].$$

This set represents all Gaussian distributions whose covariance lies within distance γ (in the Bures metric) from the nominal covariance Σ_{nom}.

Sampling strategies. Two simulation modes are available:

1. **Independent (iid) disturbances:**

$$w_t \sim \mathcal{N}(0, \Sigma_{\text{nom}}), \quad t = 0, \ldots, T-1,$$

   corresponding to temporally uncorrelated samples inside the ambiguity ball. Implementation uses a Cholesky factorization $L = \operatorname{chol}(\Sigma_{\text{nom}})$ and generates $w_t = L\, z_t,\ z_t \sim \mathcal{N}(0, I)$.

2. **Correlated disturbances:** To allow temporal correlation while preserving the same marginal covariance, an autoregressive model of order one (AR(1)) is simulated:

$$w_{t+1} = \rho\, w_t + \varepsilon_t, \qquad \varepsilon_t \sim \mathcal{N}(0, Q),$$

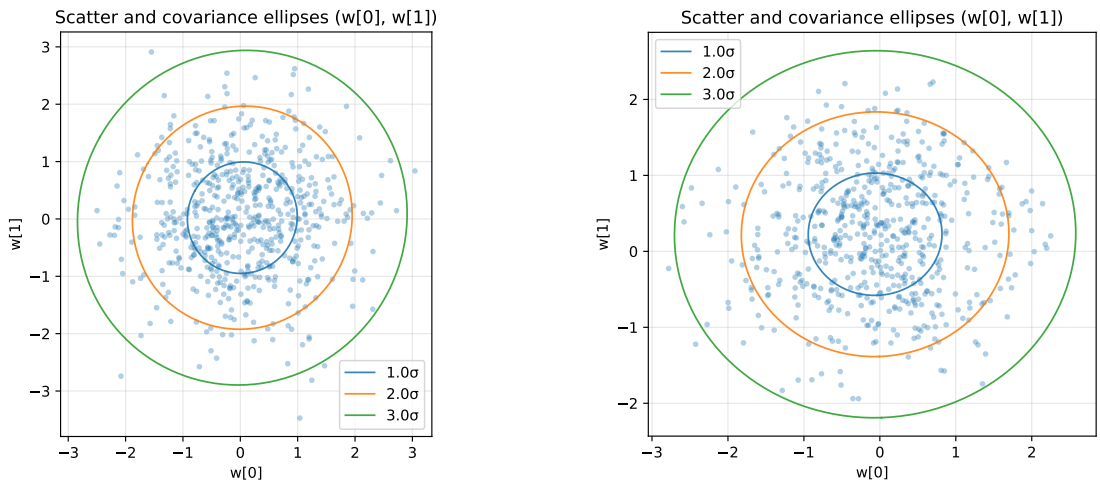
   with $0 < \rho < 1$ ensuring stability. The matrix Q is chosen so that

$$\Sigma_{\text{nom}} = \rho^2 \Sigma_{\text{nom}} + Q \quad \Rightarrow \quad Q = (1 - \rho^2)\Sigma_{\text{nom}}.$$

   This yields stationary correlated samples having marginal covariance Σ_{nom}, ensuring consistency with the ambiguity model.

In summary, the disturbances in simulation are not arbitrary noise sequences but samples from a *Wasserstein-consistent* uncertainty model. This maintains coherence between the data-driven identification, the LMI synthesis, and the subsequent Monte Carlo evaluation of closed-loop robustness.

Each figure below compares the *independent (iid)* and the *correlated (AR(1))* disturbance samplers that both lie in a 2-Wasserstein ball of radius γ around $\mathcal{N}(0, \Sigma_{\text{nom}})$. The iid and AR(1) samplers share the same stationary marginal $\mathcal{N}(0, \Sigma_{\text{nom}})$; they differ only in temporal structure (autocorrelation).



(a) Independent (iid): two-dimensional marginal samples (w_i, w_j) with nominal and empirical covariance ellipses. The empirical ellipse matches the nominal, confirming $\text{Cov}(w_t) = \Sigma_{\text{nom}}$ under iid sampling.

(b) Correlated (AR(1)): same marginal visualization. Marginals agree with Σ_{nom}; differences from iid are temporal, not marginal.

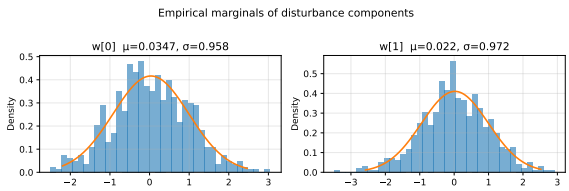
Figure A.1: Scatter and covariance ellipses. The panels visualize *marginal* covariance in the plane (w_i, w_j).

How iid vs correlated differ (and how they don't). Both modes satisfy $\mathbb{E}[w_t] = 0$ and $\text{Cov}(w_t) = \Sigma_{\text{nom}}$, hence the nearly identical scatter/ellipse and PDF panels. The difference is temporal:

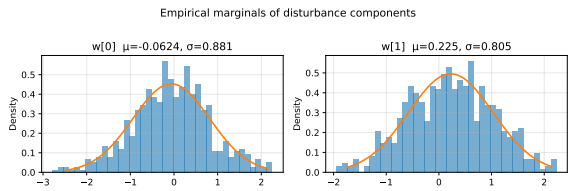
$$\text{iid:} \quad \text{Cov}(w_{t+k}, w_t) = 0 \ (k \neq 0), \qquad \text{AR(1):} \quad \text{Cov}(w_{t+k}, w_t) = \rho^{|k|}\Sigma_{\text{nom}}.$$

This is invisible in time-agnostic marginals. To expose it, add time-series diagnostics such as a trace of $w_i(t)$, an ACF plot (iid: spike at lag 0; AR(1): geometric decay ρ^k), or a lag-1 scatter (w_t, w_{t+1}) that aligns along the 45° line when correlated.

Technical caveat (boundary ellipses). The boundary covariances are constructed by eigenvalue shifts along principal axes of Σ_{nom}. This is exact in the commuting case (shared eigenvectors) and remains a useful visual surrogate otherwise; it illustrates the scale of admissible covariance perturbations at radius γ in the Bures/W_2 metric.

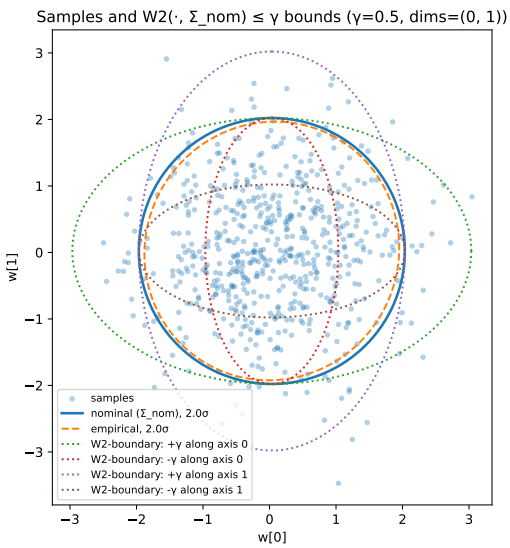


(a) Independent (iid): empirical marginal densities of disturbance components. Means are near zero and variances match the diagonal of Σ_{nom}.

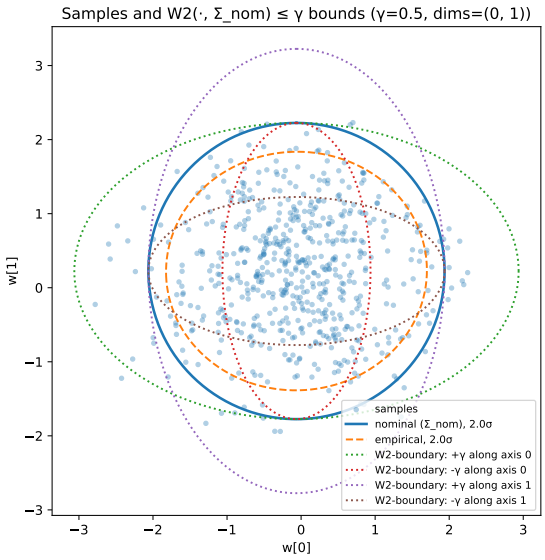


(b) Correlated (AR(1)): empirical marginal densities. As expected, they match the iid case; correlation does not alter stationary marginals.

Figure A.2: One-step marginal PDFs. Time order is ignored, so iid and AR(1) look nearly identical at the marginal level.



(a) Independent (iid): samples with nominal, empirical, and representative W_2-boundary ellipses. The nominal and empirical ellipses lie within the ball $W_2(\cdot, \Sigma_{\text{nom}}) \leq \gamma$.



(b) Correlated (AR(1)): correlated samples with the same W_2-boundary overlays. Correlation affects lag structure, not the W_2-marginal bounds.

Figure A.3: Wasserstein-ball overlays. Dotted ellipses illustrate covariances on the W_2 boundary around Σ_{nom}. They bound admissible *marginal* covariances, not individual points.

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